Advanced Programming Practice · 21CSC203P

Unit 1 · Session 6 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 6 : SLO – 1 | Procedural Concepts

Activity (Solutions)

• 1) Procedural factorial: static int factorial(int n){ int f = 1; for (int i = 1; i <= n; i++) f *= i; return f; } called as factorial(5) → 120.

• 2) Procedure vs method: a procedure is a standalone block of code performing a task (may not return a value, not tied to an object); a method is a function defined inside a class, tied to a class/object, invoked on an instance or class — every method is a procedure, but in Java all procedures must be methods of some class.

• 3) Drawbacks of procedural programming (long function demo): a 200-line main doing input, validation, computation and output mixes responsibilities — drawbacks: poor reusability (copy-paste), hard debugging, global/shared data prone to side effects, difficult maintenance and testing, no encapsulation.